



Confirmation of Candidature Review Report

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A STATISTICAL MODEL FOR THE PERSISTENCE OF THREATENED BIRDS DEVELOPED USING CITIZEN SCIENCE DATA

SHORT TITLE: PERSISTENCE MODELS FOR THREATENED BIRDS

ABSTRACT

Background

Due to certain risk factors such as climate change, habitat destruction, overhunting and pollution bird extinctions are now occurring at a rate that far exceeds their speciation rate. When a bird species is lost, the benefits it might have afforded are gone forever. There are no robust indicators of biodiversity conservation that can be used to complement existing national indicators of economic and social health. This study reviews the statistical methods that are used to model and evaluate the persistence or extinction risk of threatened bird species using citizen science data.

Methods

Adhering to PRISMA guidelines, this review systematically searched for relevant journal articles which were published between January 1900 and January 2019 in any of three databases (ProQuest Central, Scopus and Web of Science). Only the journal articles which used a statistical model, predictive model or trend analysis in analysing persistence or extinction risk (while considering risk factors) of threatened bird species developed using citizen science data were included in this study. The bird species that may be declining in population/range even in near threatened or least concern categories also included since these may be the next wave of species to be added to the endangered species lists. Citizen science data helps to increase the number of records and understanding of dynamics in declining bird species populations.

Results

This systematic review was able to identify 37 unique articles describing statistical models for this purpose. Generalised linear models and linear mixed/hierarchical models were the most popular methods used for analysis, followed by machine learning models. The review suggested several strategies to measure the persistence of threatened bird species, but there was no attempt to identify critical tipping points using methods such as change-point analysis.

Conclusion

Overall, it appears that the persistence of threatened bird species varies depending on various risk factors. Statistical models can provide a better understanding of the impact of these risk factors, which is expected to produce better immediate and long-term outcomes for the conservation of threatened birds. It is hoped that this review will identify the statistical methods which are most suitable for this purpose.

KEYWORDS

Comparative analysis, Conservation, Modelling (Population Ecology), Statistics, Species interactions

INTRODUCTION

About 10 years ago, two scientists decided to estimate the total number of birds on the planet. The number they came up with was 200 to 400 billion individual birds. Since the 1500s, birds have been declining both in terms of species numbers (by about 500 species) and numbers of individuals (by about 20-25%) (How big is the bird population? | AMNH. (n.d.)). Due to risk factors such as climate change, habitat destruction, changes to wetlands and coastal environments, predation by feral predators, overhunting, toxic pollution and ecological change (whether these are caused by natural or man-made influences), bird extinctions are now occurring at a rate that far exceeds their speciation rate. Each extinction diminishes the diversity and complexity of life on earth (Fws.gov., 2018).

The world's tropical forest bird population is estimated to be permanently reduced by 144 million individuals per year. Of all the known bird species, 129 are extinct and out of the remaining 9,787 1,313 bird species are threatened with extinction according to BirdLife International's assessment for the 2012 IUCN Red List. Of these, 197 are considered Critically Endangered and are on the brink of extinction (Butchart, S., 2013). By the year 2100, up to 14% of all bird species may become extinct, and 7-25% (28-56% on oceanic islands) may become "functionally extinct", in the sense that the number of individuals within these species will become so low that the species will no longer play an important role in their ecosystem (How big is the bird population? | AMNH. (n.d.)).

Birds play a vital role in ecosystems. For example, birds pollinate and disperse the seeds of many plants, including many crops important to humans. They are also helpful in controlling crop pests. Scavenger birds such as vultures help with the decomposition of organic material (How big is the bird population? | AMNH. (n.d.)). When a bird species is lost, the benefits it might have afforded are gone forever (Fws.gov., 2018). All of these benefits to humans and other organisms will likely decline along with bird numbers and species. Because birds can have an important role in controlling other organisms, their decline may even encourage the spread of disease (How big is the bird population? | AMNH. (n.d.)).

Persistence of species has been defined as "*the continued or prolonged existence of species*". The past decade has seen a resurgence of interest in extinction, yet research on this topic is still at a reconnaissance level, and the present understanding of its role in evolution is weak. There are no robust indicators of biodiversity conservation that can be used to complement existing national indicators of economic and social health (Raup, 2018).

Citizen Science (CS) can directly contribute to biogeographical knowledge and conservation policies by increasing the number of recorded species in large geographic areas. The cost-effectiveness of CS data collection offers the potential for scientists to tackle research questions on large spatial or temporal scales (Aceves-Bueno et al., 2018). The PPWCMA Annual report 2019 provides an example of the power of citizen science for monitoring the persistence of birdlife in the vicinity of an Australian city (Melbourne). Between the 2005-06 and 2016 analyses the number of wildlife sightings increased from 437,845 to over 3 million due to a proliferation of citizen science survey programs (CMA, 2019).

In this systematic review the statistical modelling techniques used to identify suitable factors to explain and measure persistence or extinction risk of threatened birds will be considered. Most of the studies have considered the risk factors of habitat destruction and climate change and all the studies have analysed citizen science database or survey data while focusing on specific breeding and non-breeding seasons, predicting bird species behaviour adaptation by utilizing various kinds of statistical model.

However, in this systematic review we focus only on the journal articles that consider threatened bird species under IUCN red list assessment criteria, and the bird species that may be declining in population/range (in near threatened or least concern categories) as these may be the next wave of species to be added to the endangered species lists. These articles consider persistence or extinction risk measures, which are collected through citizen science data and used in some form of quantitative analysis using statistical modelling, predictive modelling or trend analysis to assess the persistence of threatened bird species.

It is expected that this review will help researchers to identify appropriate statistical models for evaluating persistence measures for identifying threatened bird species worldwide using citizen science data. Further we hope that the findings of this systematic review will help to create more effective and efficient responses for saving threatened bird species worldwide in the future.

MATERIALS AND METHODS

Registration and protocol

This systematic review was reported in accordance with the preferred reporting items for systematic review and meta-analysis protocols (PRISMA-P) 2015 statement (Moher D et al., 2015). The protocol for this review has been published in protocols.io and osf.io in order to enhance the reproducibility of the results. The protocol can be accessed respectively from: <http://dx.doi.org/10.17504/protocols.io.yqwfvx> and <http://dx.doi.org/10.17605/OSF.IO/6VGUK>.

Literature search strategy and assessment

The study considered persistence or extinction risk of threatened birds conducted using citizen science data. To find the relevant research articles, an individualized search strategy was built for the Web of Science, Scopus and ProQuest Central databases. This literature search was limited to the journal articles which were published from January 1900 to January 2019. The above databases all show an increasing trend in research focused on threatened bird species from 1990 onwards because of the increasing use of citizen science data. Because of time constraints it was decided to include only articles published in English. Only individualized keywords were used in the search strategy for the above-mentioned databases, in order to extract more relevant articles, and only defined key words were used because these databases do not have a library indexed language. The following list denotes the series of key words used for each database.

Web of Science

(TS=("bird*") AND TS=("threatened" OR "endangered" OR "vulnerable") AND TS=("citizen science") AND TS=("persistence" OR "extinction risk") AND TS=("statistic*" OR "statistical model*" OR quantitative OR "Models, Statistical" OR trend* OR "predictive model*")) AND LANGUAGE: (English)

Scopus

("bird*") AND ("threatened" OR "endangered" OR "vulnerable") AND ("citizen science") AND ("persistence" OR "extinction risk") AND ("statistic*" OR "statistical model*" OR "Models,Statistical" OR "trend*" OR "predictive model*") AND (LIMIT-TO (SRCTYPE , "j")) AND (LIMIT TO (LANGUAGE , "English"))

ProQuest Central

((bird*) AND ((threatened) OR (endangered) OR (vulnerable)) AND (citizen science) AND ((persistence) OR (extinction risk)) AND ((statistic*) OR (statistical model*) OR (Models, Statistical) OR (trend*) OR (predictive model*))) AND la.exact("ENG") AND PEER(yes) AND stype.exact("Scholarly Journals")

Inclusion and exclusion criteria

To select articles better matched to the objective of this systematic review, four main inclusion criteria were developed. Articles were selected if (i) the study focused on threatened bird species under IUCN red list assessment criteria, and, for bird species in near threatened or least concern categories that are declining in population/range, because these may be the next wave of species to be added to the endangered species lists; (ii) Citizen Science data were used to conduct the study; (iii) the study discussed persistence or extinction risk of threatened bird species; (iv) the study needed to carry out a quantitative analysis using statistical modelling or predictive modelling or trend analysis.

Selection process and data extraction

The initial database search was able to identify a total of 2336 articles which included 2158 articles from ProQuest Central, 177 articles from Scopus and 1 article from Web of Science. A manual search added 21 articles not included in these three databases. These articles were uploaded to EndNote software and 87 duplicates were removed by the first author. The other two authors were then given online access to view the results. The first two authors then independently screened the titles and abstracts against the inclusion criteria and shortlisted the most relevant articles. The articles which had been accepted by both reviewers were shortlisted and the articles which had questionable eligibility were directed to the third reviewer for resolution. This process shortlisted 55 articles for a full text review, out of which the team agreed on 37 unique articles for inclusion in this review. The following PRISMA flow chart (Fig 1) outlines the above search and review process.

Quality Assessment

Since there is no quality assessment checklist for journal articles in this area a new checklist named “Udani DM Checklist” was created, referring to the Downs and Black checklist (Black, N., 1998), in order to assess the quality of these 37 articles. This checklist consists of 22 items distributed among five sub scales, namely; reporting (9 items), external validity (4 items), internal validity - bias (4 items), internal validity - confounding (4 items) and power (1 item). The performance of the quality assessment items is assessed using “1” for compliance and “0” for non-compliance. The standard deviation for each item is calculated as a check of the discrimination capability for each item. The maximum score that a study can obtain is 22, with higher scores indicating better quality. In this review the Udani DM scores have been categorized as follows; low quality (10-13), fair quality (14-15), average quality (16-18) and high quality (19-22). Table S1 shows the 22 questions included in “Udani DM Checklist”.

Taxonomy for Journal Articles

There are many different statistical techniques that can be used to evaluate bird persistence measures using citizen science data. The “guide for choosing among statistical techniques” published in (Tabachnick & Fidell, 2013) has been used to classify the various techniques found in the shortlisted articles. These articles have also been classified in terms of the type of citizen science data used, the type of persistence data used, and the nature of the risk factors identified.

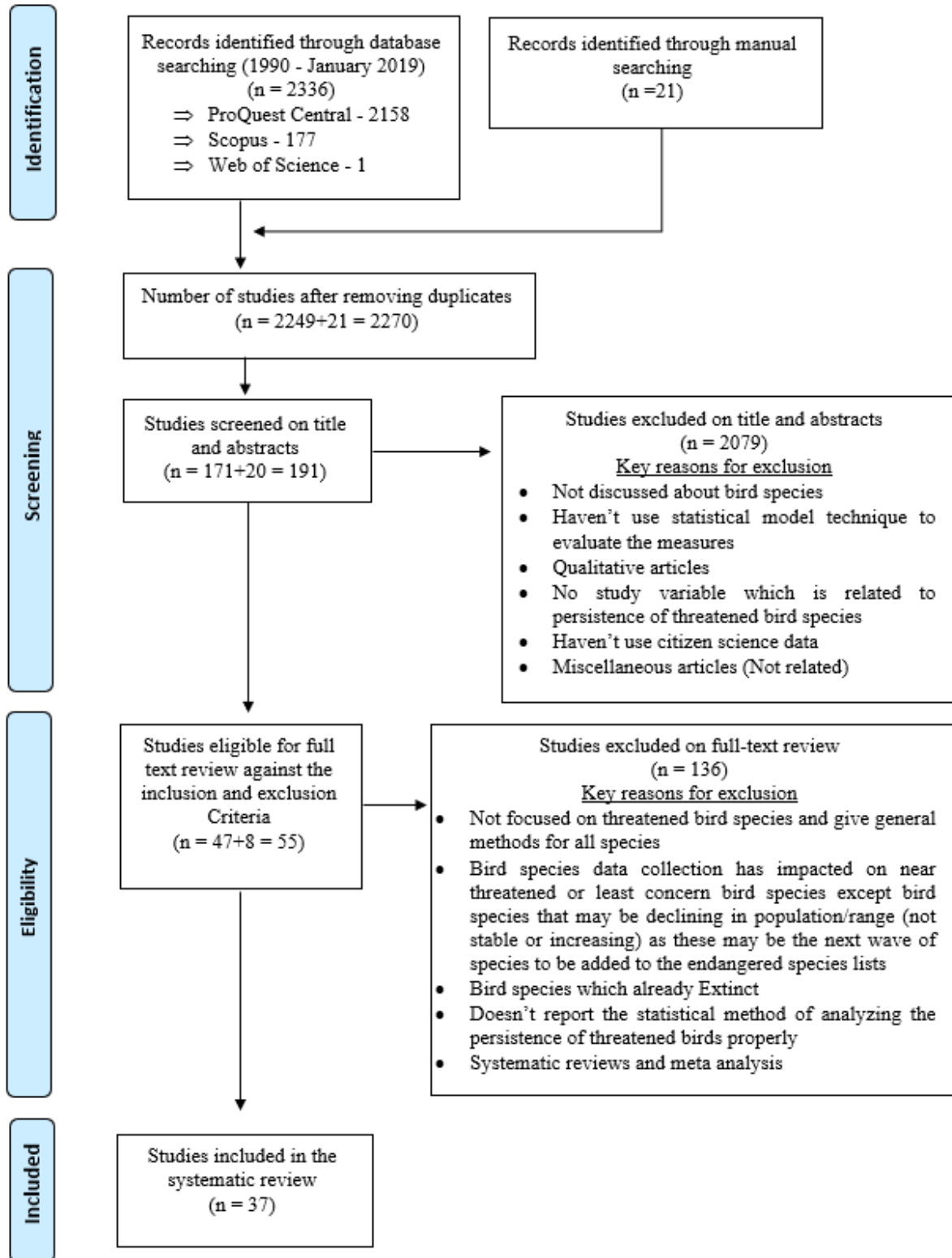


Fig 1. PRISMA flow diagram.

RESULTS

This review refers to studies conducted between 1727 to 2017 in 10 different countries, with more than one country providing the data for some of the articles (Boakes, Isaac, Fuller, Mace & McGowan 2018; Wilson, Anderson, Wilson, Bertram & Arcese 2013; Studds et al., 2017; Yong et al., 2017). The highest number of studies was carried out in the United States of America (12) and Africa (7), followed by the East Asian–Australasian Flyway (3), Canada (3), Australia (2). Due to the variability in the included studies in terms of study design, sample size, persistence or extinction risk, objectives and statistical methods for analysing the data, this systematic review will provide a qualitative review, rather than a meta-analysis.

Approximately half (49%) of studies were discussed only a single bird species such as Carnaby's cockatoo (*Calyptorhynchus latirostris*), Aransas-Wood Buffalo Whooping cranes (*Grus americana*), Western Grebes (*Aechmophorus occidentalis*), Black Harrier (*Circus maurus*), Western capercaillie (*Tetrao urogallus*). Group studies of species were used for the remaining studies; for example, forest pigeons, North American quails, breeding species of Chinese protected birds, most abundant migratory shorebirds.

In these 37 papers, citizen science data were analysed using more than 10 different statistical models and these models can be categorized into four categories. A summary of the statistical methods grouped under these categories are listed below.

- (a) Generalised linear models (32%) including General linear models, Generalised additive models, Poisson regression models, ANOVA, logistic regression models and least square regression model (Pearse et al., 2018; Erickson, Found-Jackson & Boyce 2017; Dhanjal-Adams et al., 2019; Carvalho et al., 2014; Downs, Pfeiffer & Hart 2014; Pryke, Samways & Hockey 2010; Shaw, Jenkins, Allan & Ryan 2015; Kamp, Koshkin & Sheldon 2009; Amar & Cloete 2017; Aubry, Desrochers & Seutin 2016; Yong et al., 2017; Xu et al., 2019).
- (b) Machine learning models (24%) including predictive models, maximum entropy models, boosted regression trees (BRT), random forests (RF), clustering models (Ramesh, Gopalakrishna, Barve & Melnick 2017; Rösner, S., Brandl, Segelbacher, Lorenc, & Müller 2014; KrÓlikowska, Krupiński & Kuczyński 2017; Zlonis, Panci, Bednar, Hamady & Niemi 2017; Wogan, 2016; Tanner, Papeş, Elmore, Fuhlendorf & Davis 2017; Li et al., 2015; Hart, Kelly & Pearson 2018; Runge, Tulloch, Hammill, Possingham & Fuller 2015).
- (c) Linear mixed/Hierarchical models (22%) including general linear mixed models, mixed effects cumulative logit regression model, hierarchical mixed effect model (Garcia-Heras et al., 2019; Wilson, Anderson, Wilson, Bertram & Arcese 2013; Rushing, Dudash, Studds & Marra 2015; Berry, Benkman, Muela, Seminario & Curti 2010; Wilson, Smith, Naujokaitis-Lewis & McCarthy 2018; Studds et al., 2017; Thogmartin, Sauer & Knutson 2004; Clemens et al., 2016).
- (d) Miscellaneous (22%) including deterministic model, models for occupancy, persistence model, stage structure matrix model and count-based population viability analysis (Williams, Yates, Saunders, Dawson & Barrett 2017; Lee, Altwegg & Barnard 2017; Boakes, Isaac, Fuller, Mace & McGowan 2018; Ward et al., 2015; Kalle, Ramesh & Downs 2018; Erickson, Found-Jackson & Boyce 2014; Wootton & Bell, 2014; Culp, Cohen, Scarpignato, Thogmartin, & Marra 2017).

Out of these 37 papers only one paper used a combination of the above categories, namely generalised linear and machine learning models (Hart, Kelly & Pearson 2018). Since this paper mostly used machine learning techniques, it is included in group (b). Table S2 summarizes the information extracted from the shortlisted articles.

Monte Chain Monte Carlo (MCMC) estimations were used in 19% of analyses to fit the models. Akaike's Information Criterion (AIC) or Area Under Curve (AUC) were used to identify the best model in 46% of analyses. The majority of statistical analysis were conducted using R software, followed by JAGS in a Gibbs-sampling environment, and MATLAB. All the machine learning models were generated using Maxent software with only one exception (Ramesh, Gopalakrishna, Barve & Melnick 2017). Maxent software is used for the modelling of species niches and distributions determined from a set of environmental or climate layers together with a set of sample locations where the species of interest has been observed. This tool compares well with or outcompetes other modelling technique and it has been shown to be similar to more conventional regression-based approaches used for modelling environmental correlates. Maxent can be applied to address a variety of ecological questions depending on how the models are calibrated and evaluated.

All the studies used observational data collection designs where 49% of studies were conducted using citizen science databases such as; eBird, BirdLife Australia, Southern African Bird Atlas Projects (SABAP 1 and 2) and Christmas Bird Count Data. The remaining 51% of studies were conducted using citizen science survey data to conduct their study of occurrence in protected areas, forests, country-side or urban areas. Survey data were collected using several methods such as; opportunistic sighting and telemetry data (Pearse et al., 2018), marked with GPS-GSM or PTT tracker devices (Garcia-Heras et al., 2019), fresh capercaillie droppings (Rösner, S., Brandl, Segelbacher, Lorenc, & Müller 2014), stable hydrogen isotopes (Rushing, Dudash, Studds & Marra 2015), pitfall trapping (Pryke, Samways & Hockey 2010) etc. Researchers considered important factors affecting species trends such as breeding or non-breeding season, climate suitability, age and sex, seasonal changes and human population density near the area while collecting the occurrence data. There are two types of data collected for these studies categorized as abundance, which tracked the total population size, and presence-absence surveys, which tracks the area of occupancy (Joseph, Field, Wilcox & Possingham 2019). The majority of the studies (81%) were used presence-absence surveys for their analysis whereas (19%) of the studies used abundance surveys.

In this review, risk factors that may influence the persistence of threatened bird species were categorized as biotic and abiotic. The biotic risk factors include habitat loss, land use change, climate change, prey availability, long migration distance and predation. The abiotic risk factors are attributed to human influence factors such as hunting, agriculture intensification, additional recreational activities, illegal pet trade, high collision mortality on power lines. Habitat loss was the most commonly cited risk factor (32%), followed by abiotic risk factors (24%), climate change (23%) and land use change (11%). Tables 1 show a summary of the statistical methods grouped under these categories. Out of these 37 papers 34 also discussed the conservation efforts being made to protect threatened bird species.

Quality assessment

A quality assessment was carried out for all the 37 studies using the Udani DM quality assessment tool. The quality scores ranged between 10 and 20 with a mean score of 16. Three studies were categorized as high quality, 11 studies were categorized as fair quality, 19 studies were categorized as average quality and 3 studies as low quality. The standard deviation for

each item in the checklist was calculated to identify items with greater discriminatory power in terms of quality.

Table 1. Statistical methods categorization

Category		Methods				Total
		(a)	(b)	(c)	(d)	
Citizen science	Database data	4	6	3	5	18
	Survey data	8	3	5	3	19
Nature of survey data	Abundance data	4	0	3	0	7
	Presence-absence data	8	9	5	8	30
Risk factors	Biotic	18	13	14	11	56
	Abiotic	6	3	4	5	18

* Total number of risk factors mentioned

Limitations

There are some limitations for this systematic review. In this systematic review, we only considered studies which related to the persistence of threatened birds which used citizen science data. In addition, the review was limited to peer reviewed articles published in English and only to articles that performed a quantitative analysis using a statistical modelling technique. Finally, a meta-analysis has not been performed due to the wide range of statistical methods applied, the wide range of research objectives and species studied.

DISCUSSION

In this review we identified three main categories of statistical approaches for describing persistence data obtained through citizen science; namely general linear models used for cross-sectional studies, machine learning models used for large distributional studies, linear mixed/hierarchical models used for spatial and temporal multi-level studies. The most popular method of analysis is generalized linear models, including ANOVA, logistic regression model and least square regression models, followed by machine learning models including predictive models, maximum entropy models, boosted regression trees (BRT), random forests (RF) and clustering models. Linear mixed/hierarchical models are the third most preferred method including general linear mixed models, mixed effects cumulative logit regression model, hierarchical mixed effect models. Other models categorized into miscellaneous group consists mainly models of occupancy for persistence.

Generalized linear models commonly used to tested for changes in occupancy over time or to evaluate population trends. Machine learning models such as maximum entropy models used to test which variable selection approach performed best for creating ecological niche models and predictive models used to predict the distribution of each species from the occurrence data sets. All these models were adopted using Maxent. All hierarchical models over-dispersed Poisson variables and models were fit using Markov chain Monte Carlo methods. Hierarchical Bayesian approaches were used in all cases since they are well suited for population time series analyses because they provide a robust framework to account for sources of variation at multiple scales including observer effects, overdispersion and the spatially specific effects of drought across the species range (Thogmartin, Sauer & Knutson 2004). And the mixed models used highly to quantify the effects of the predictors. In miscellaneous models, mainly used dynamic occupancy models were applied to evaluate how the covariates were linked to colonization of unoccupied sites and extinction of occupied sites by bird species.

According to the studies in this review, all the studies are observational where the citizen scientists collected occurrence data of threatened bird species, mostly using survey data. Modern technology is providing many exciting new methods for collecting this survey data. Citizen science databases including eBird, BirdLife Australia, Southern African Bird Atlas Projects (SABAP 1 and 2) and Christmas Bird Count Data were collected by private individuals. Individual volunteer accuracy varies, depending on task difficulty and volunteer experience; multiple methods exist for boosting accuracy to required levels for a given project. Data produced through citizen science may contain error and bias and can be mitigated using existing statistical tools. Many of the systematic biases in citizen-science data are the same biases that occur in professionally collected data such as; spatially and temporally non-random observations, non-standardized capture or search effort, under-detection of organisms, confusion between similar-looking species and the over-or under-reporting of rare, cryptic, or elusive species as compared to more common ones. The only known bias specific to citizen science is the potentially high variability. The methods like expert scrutiny of images provided, standardization of measurement tools and collection of instrument calibration data and collect multiple independent volunteer measurements of each subject could be used to improve the quality of this data (Kamp, Koshkin & Sheldon 2009).

For survey data most of the studies used presence-absence data rather than abundance data. Because at low budgets, the estimated decline rate from presence-absence data was reasonably good, and the variation was much lower than that obtained from abundance data (Joseph, Field, Wilcox & Possingham 2019). Counts provided by citizen science data can be regarded as indices to population size and used to estimate spatial and temporal patterns in relative population abundance (Link and Sauer 1998).

About the analysis of risk factors indicate that many species have experienced range contractions due to biotic risk factors including habitat loss, climate change, land use change, prey-availability, long migration distance and predation. In addition, abiotic risk factors including hunting, agriculture intensification, increment of recreational activities, illegal pet trade, high collision mortality on power lines have been identified as risk factors. While quantifying these risk factors the studies also considered other factors such as; breeding and non-breeding habitat quality, breeding and non-breeding territory quality, age and sex, human population density near the species area, the estimated size of the area, the latitude and longitude of each site. Environmental data such as rainfall, temperature, snow cover and mapping data were obtained using databases such as WorldClim and the NOAA Earth System Research Laboratory. Some of the studies generated predictive habitat maps (ecological niche modelling) from bioclimatic, topographic, and land-use variables derived from distribution record, allowing the examination of trends in species' distribution using Maxent software (Berry, Benkman, Muela, Seminario & Curti 2010).

As a general rule in ecology, occupancy is correlated with abundance (Andrewartha & Birch, 1954) which means that occupancy surveys can be used to predict abundance for a variety of applications. However, in the context of a threatened species, it is important to examine occupancy and abundance separately, because a high probability of persistence might be the result of a historically large bird species colony rather than a reflection of current abundance.

The actions needed to help save most threatened species are often not clear and for many species research is required to enhance recovery efforts. The most urgent actions are to protect and effectively manage Important Bird and Biodiversity Areas, control or eradicate invasive alien species, restore habitat, control hunting, raise awareness, and, in some cases, establish captive breeding and reintroduction programmes. There is good evidence that such actions can

bring species back from the brink of extinction, with a suite of species that were once reduced to tiny populations now well on the road to recovery (Butchart, 2013).

Conservation planning activities generally require knowing what and where habitats need to be protected in order for bird species populations to be sustainable. Population monitoring is therefore key to the effective conservation of threatened species. Because species' range and distribution analysis allow the evaluation of impacts and responses to risk factors such as climate change and land use, statistical modelling has an important role to play in conservation. These models are also important for the identification and determination of exposure levels of potential threats for individual species. Studies examining life history traits are also needed for species that are of high conservation concern and experiencing elevated rates of extinction. The capability to model and map abundances of high priority threatened bird species over large regions is therefore critical for the identification and prioritizing of habitats for conservation action and for future monitoring.

CONCLUSIONS

In summary, this systematic review was able to identify three main classes of statistical modelling methods that have been used to analyse occurrence data obtained using citizen science databases and surveys. They are general linear models, machine learning models/hindcast models and linear mixed/hierarchical models. It has been found that occurrence data behaviour depends on risk factors. Therefore, these risk factors should be controlled or even minimize the effect when gathering occurrence data to assess program outcomes.

Contrary to expectation no studies with change point analysis to analyse species occurrence using citizen science data were found. Change-point analysis is a powerful new tool to determine animal movement data which characterizes the changes, controls the overall error rate, robust to outliers, more flexible and is simpler to use especially when analysing large data sets. Analysis of movement data is straightforward because the data are multidimensional and autocorrelated in space and time. Random walk models were the next ended to several mathematically and statistically sophisticated Bayesian approaches to interpret movement data such as State-Space Models (SSM) and Brownian Bridge Movement Model (BBMM) which can help understanding the under lying process driving animal movement and appeared successful in estimating animal paths, migration routes and analysing fine-scale resource selection. Still change point analysis was tested for data obtained using such as; GPS data loggers, GSM tracking devices, satellite tracking devices. These methods can better to build a persistence measure using citizen science data for the serial autocorrelation in long occurrence data series allowing unbiased estimation.

In addition, the effect of individualized citizen science data on long term species occurrence behaviour should also be tested in future studies.

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SUPPORTING INFORMATION

Table S1: Udani DM Checklist

Table S2: Summarized table